

Element B-

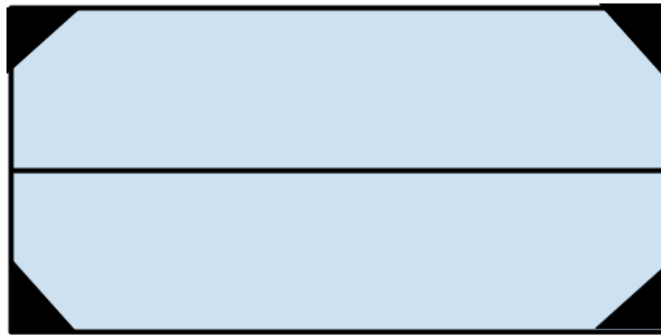
Overview:

This class at Brentwood High School has been working on this “zoo animal lift project” for the past two years now. The first year failed to fulfill its intended purpose, which is carrying and moving a partially sedated animal. While the second year group did improve upon the previous design, they yet again fell short of the intended purpose. We have the past two groups projects available to us for us to look at and examine, putting us at a comparative advantage. We are able to build upon the previous years successes and failures, as well as consult with experts at the zoo over how we can better address problems such as the integrity of the frame and balancing issues.

2 Years Ago

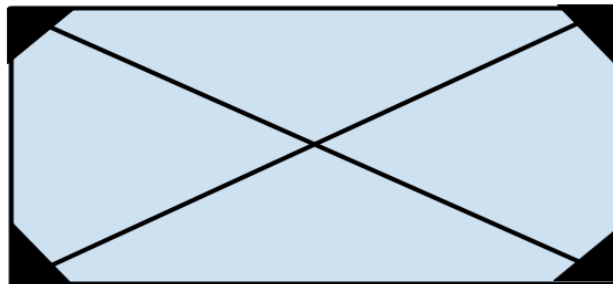
2 years ago was the first attempt at making the animal lift. While it had many flaws, it laid the framework for the project for the years to come. This design was straightforward, and also very flimsy. They used a 4x6 ft metal frame made of 4 metal beams as the base, in which 6 ratchet straps descended from the frame to hold the tarp. Cables hung from the engine holding the frame into place. One strength they had was the implementation of a beam down the middle of the frame to acknowledge the need for better stability, and while it did help, it was not enough stability for the intended purpose. The frame is made out of galvanized steel unistrut. The bracing of the corners was only done with duct tape; according to the New York Times duct tape can only support up to 180 lbs when triple layered. The goal is to have the frame prepared to

carry an animal up to a ton. There are also problems with the use of a single I-beam to brace a rectangular frame. According to Northern Weldarc, a welding company in Canada, these problems include lack of resistance to horizontal forces and buckling. The lack of resistance to horizontal forces would be the tendency to sway easier if weight in the hammock was shifted. The I-beam provides little to no security for this. An I-beam also buckles under a lower than ideal amount of weight. We are expected to carry 1500-2000 lbs and this frame began to bend over the weight of a single human.



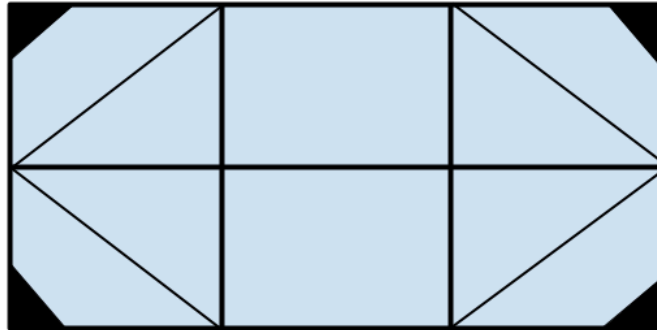
1 Year Ago

This design was a big improvement from the last one, and even though it didn't solve the problem, it came closer to closing the gap. Some changes they made include the implementation of an X-beam, corner bracing, and most importantly, the use of an engine leveler. The engine leveler was a great idea, but it was not easily adjustable unless you cranked it to adjust manually. First of all, an X-beam makes the frame far more durable and able to carry heavier loads. The X-beams improved performance includes protection against horizontal forces, stability, and load distribution. It was better at addressing sway and tilt, seeing as it wasn't buckling as much on the sides of the frame. That being said, it still was buckling at the weight of just one person on the tarp, furthermore proving that integrity was the biggest issue.



This Year

Our goal for this year's project is to finally finish the Zoo animal lift project. We have a budget of \$225, as well as receiving the materials from the years prior. We loved the idea of using an engine leveler in order to sustain side-to-side stability. We realized early on that one engine leveler would not be enough, and the hardships with adjusting them rendered them useless. We had to idea to incorporate a switch to easily adjust both engine levelers simultaneously in real time. The code to do this was written by our peer Joe Johnson. The way we are able to have two engine levelers is by having two points of contact along the track. This includes the original engine as well as another set of wheels attached to the engine by a shepherd's crook that is following behind. We also completely redesigned the frame with Corbin Wolfgang's welding expertise. He reattached steel beams at different points on the frame, making a coffin-style steel frame. He also made the corner bracing uniform on each corner to avoid diagonal buckling. We lastly spray-painted this frame black in order to prevent rust as well as look cleaner and better. This newly reinforced frame completely addresses the integrity problem being faced; it can adequately carry the required load of 1500-2000 lbs now.



Helicopter Lift

A Helicopter cargo lift uses a similar frame below the bottom of the helicopter in order to stabilize the large loads being carried while traveling at high speeds. Helicopter lifts take many factors into account, including rigging, load positioning, and sway reduction. At first glance, a helicopter lift seems much different from our project, the concept is very similar. The biggest difference is that the setup of the procedure of carrying a load by helicopter has to be a lot more detail-oriented. The stakes are a lot higher, seeing as the worst case scenario is dropping and losing the load entirely, but also dropping a load somewhere where it shouldn't be and harming somebody or even

the environment. Our worst case scenario is our design failing and not being used at the zoo, causing another delay for the animal lift to students next year.

According to Fair Lifts Helicopter service, rigging a helicopter lift is vital to making sure the load is stable and supported, and furthermore makes sure the center of gravity is stable and in one place. An unbalanced load can make it hard to pilot the aircraft and potentially make it dangerous for anybody in the helicopter. Another reason rigging is important is to prevent any load swing or the pendulum effect. If a load is improperly rigged, it can start to swing heavily or spin while in the air. This is also dangerous to any passengers. Rigging techniques involve using sturdy taglines and multiple attachment points to maintain control. We are implementing multiple attachment points by using two engine levelers as opposed to only one last year, and zero the year before. We are also adding another point of contact trailing behind the main engine on the track to support our implementation of a second engine leveler. Another way we use rigging is to ensure efficiency; a helicopter's fuel is not infinite, and adequate rigging makes sure the aircraft can get from point A to point B as quickly as possible. Many loads attached to helicopters also have a quick-drop mechanism, which is used in times of emergency. This specialized equipment is incorporated when rigging the load onto the helicopter. Taking all of these factors into account, improper rigging can lead to hazardous and unpredictable scenarios. Rigging prevents disaster by ensuring the load is stable, and in our case, proper rigging can keep an animal on our hammock and out of harm's way.



Crane Lift

Much like a helicopter cargo lift, a crane lift relies heavily on precise rigging to ensure the stability and safety of the load being moved. While the environments differ, cranes have a much more fixed environment than a helicopter. Regardless of the conditions, load management remains the same. One of, if not the most critical part of load management is the center of gravity, or rather COG. If the crane's hook isn't positioned over the center of gravity, this can pose serious problems like swinging and tilting while suspended. Instability endangers the expensive equipment but more importantly puts people at risk. To mitigate this, riggers like to perform test lifts because adjusting the rigging is necessary to fully achieve a balanced load. Rigging techniques are very important in maintaining load stability. By using multiple attachment points a load's COG will have less negative impact with multiple slings. Additionally, the use of taglines, or ropes attached to the load, allows ground personnel to control and guide the load during movement, reducing the risk of collision with obstacles or personnel. In our project, we have taken these principles into account by implementing multiple attachment points using two engine levelers, an improvement from previous iterations. Our two engine levelers are easily adjustable by the use of tandem electric motors that can balance the lift simultaneously. We implemented another engine leveler based on helicopter and crane lifts as well as the basic concept that addresses the tilt of forward and backwards by having multiple attachment points.

Safety protocols also emphasize the importance of pre-lift inspections and planning. Before any lift, it's vital to inspect your rigging for damage and make sure that

the path is clear. Moreover, understanding potential hazards allows for the development of contingency plans, such as having a quick-release mechanism in place for emergencies. While the consequences of a crane lift failure may differ from those of a helicopter lift, the underlying risks to equipment, personnel, and project timelines are significant. That being said, it is important to be very mindful and precise when preparing a crane lift; for our case we need to be mindful since we are potentially dealing with the lives of animals.

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